

Technology #05

Phage Anti-Biofilm Coating for Shunt
Infection Prevention

Engineering Technology-Transfer Dossier
CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION
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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Phage Anti-Biofilm Coating for Shunt Infection Prevention

WHY DOES IT MATTER?

Infection causes 8-12% of shunt failures at \$40-80K per case

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 4 governing equations, 11 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in Microbiology + Surface Science + Biomaterials. 7-step build plan with specific test articles and acceptance criteria. Target buyer: Anti-infection medical device company; neuroshunt manufacturer.

WHAT DOES THE BUYER GET?

Phage K + Ti coating concept + mechanism description; Phage adsorption + Monod biofilm governing equations; Bench test protocol (ASTM E2180, ISO 22196, titer assay); External precedent catalog (phage K, Ti coatings); Critical UNKNOWN disclosure (phage stability, immune response, resistance); This engineering dossier with failure analysis + build plan

WHAT DOES THE BUYER HAVE TO BUILD?

cGMP-grade phage K manufacturing process; Electrospun Ti coating process for catheter geometry; Phage immobilization chemistry + QC specs; Aseptic assembly process (terminal sterilization likely incompatible); Regulatory submission (likely combination product — biologics + device; complex pathway); Clinical validation (infection rate endpoint); Long-term stability + resistance evolution monitoring

WHAT IS THE NEXT DECISIVE EXPERIMENT?

Ti-coated catheter coupons (various electrospinning parameters)

WHAT WOULD MAKE US KILL IT?

Phage K cannot retain infectivity when immobilized on Ti surface for >24h

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 5 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 7-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: Ti-coated catheter coupons (various electrospinning parameters)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - Ti-coated catheter coupons (various electrospinning parameters) - 12 weeks

Basis: Build plan WP. Scope: Coating thickness, surface area, adhesion. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - Phage-coated Ti coupons - 10 weeks

Basis: Build plan WP. Scope: Gamma_0 (surface PFU/cm²), infectivity retention. Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - Phage-coated specimens in CSF mimic at 37°C - 12-24 weeks (chronic)

Basis: Build plan WP. Scope: Titer vs time. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 4 equations. Key assumptions: Phage K host range limited to *S. aureus* (literature-supported); Phage retains infectivity when immobilized (UNKNOWN — depends on immobilization chemistry); Biofilm establishment requires minimum bacterial colonization (literature-supported); CSF environment supports phage stability (UNKNOWN — CSF composition may affect phage); No host immune response to phages (UNKNOWN for chronic exposure). UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

5 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: Ti-coated catheter coupons (various electrospinning parameters). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: *S. aureus* phage K immobilized on electrospun titanium-coated catheter surface prevents biofilm establishment

POTENTIAL VALUE DRIVER: Addresses: Infection causes 8-12% of shunt failures at \$40-80K per case

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: Anti-infection medical device company; neuroshunt manufacturer

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: Microbiology + Surface Science + Biomaterials

Disciplines: Microbiology (phage-bacteria kinetics), Surface science (biofilm formation, anti-biofouling), Biomaterials (titanium coatings, electrospinning), Sterilization and stability (phage viability)

Architecture: CSF shunt catheter with *S. aureus* phage K immobilized on electrospun titanium-coated surface; phages prevent biofilm formation by infecting and lysing *S. aureus* before biofilm establishes

Subsystems:

SS-01: Catheter body (substrate) - Carries CSF flow + supports electrospun Ti coating

SS-02: Electrospun titanium coating - High-surface-area substrate for phage immobilization; Ti has known antibacterial properties

SS-03: Phage K immobilization layer - *S. aureus* phage K bound to Ti surface; infects and lyses *S. aureus* on contact

SS-04: Stabilization matrix for phage viability - Preserves phage infectivity over implantation lifetime (likely trehalose or lyoprotectant)

2. Mechanism Architecture

Physical Changes: *S. aureus* cells contacting phage-coated surface adsorb phages, become infected, and lyse before establishing biofilm

Key Physics: Phage adsorption kinetics: $dP/dt = -k_{ads} * [Phage_{surface}] * [Bacteria]$. At sufficient phage surface density, bacterial colonization rate < lysis rate, preventing biofilm establishment.

3. Governing Engineering Model

Summary: Phage-bacteria adsorption kinetics on immobilized phage surface. Surface phage density must exceed threshold such that bacterial lysis rate exceeds colonization rate during the critical 24-48h post-implantation window.

Equations:

$dP/dt = -k_{ads} * [Phage] * [Bacteria]$ [phage-bacteria binding kinetics]

$\mu = \mu_{max} * S / (K_s + S)$ [Monod specific growth rate]

$\Gamma_{phage}(t) = \Gamma_0 * \exp(-k_{decay} * t)$ [phage surface density decay]

Threshold: $k_{lysis} * \Gamma_{phage}(t) > k_{attach} * [Bact]_{challenge}$ for t in $[0, t_{critical}]$

Assumptions:

- Phage K host range limited to *S. aureus* (literature-supported)
- Phage retains infectivity when immobilized (UNKNOWN — depends on immobilization chemistry)
- Biofilm establishment requires minimum bacterial colonization (literature-supported)
- CSF environment supports phage stability (UNKNOWN — CSF composition may affect phage)
- No host immune response to phages (UNKNOWN for chronic exposure)

Boundary Conditions:

- Bacterial challenge: *S. aureus* at 10^2 - 10^5 CFU/mL (typical perioperative contamination)
- Surface: phage-coated Ti on catheter inner and outer lumen
- Time: 0-72h post-implantation (critical window)

Failure Regimes:

- $\Gamma_0 < \text{threshold}$ — phage density insufficient; biofilm establishes
- k_{decay} too high — phage inactivated before critical window ends
- Immobilization blocks phage receptor — phage cannot bind bacteria
- Bacterial challenge exceeds phage capacity — overwhelming infection

- Host immune response clears phages prematurely

4. Design Inputs

ID	Input	Value	Origin
DI-001	Clinical need	Infection = 8-12% of shunt failures, \$40-80K per case. Antibiotic resistance ever rising (R332)	EXTERNAL_PRECEDENT
DI-002	Functional requirement	S. aureus phage K immobilized on electrospun Ti-coated catheter surface; phage K biofilm formation	MODELLED
DI-003	Target organism	Staphylococcus aureus (primary shunt pathogen, per literature)	EXTERNAL_PRECEDENT
DI-004	Critical prophylactic window	24-48h post-implantation (infection establishment window)	EXTERNAL_PRECEDENT
DI-005	Bacterial challenge (worst case)	10^2 - 10^5 CFU/mL perioperative contamination	EXTERNAL_PRECEDENT
DI-006	Antimicrobial efficacy target	Log reduction ≥ 3 (99.9%) at 24h vs S. aureus (ASTM E2180)	MODELLED
DI-007	Biocompatibility (ISO 10993)	UNKNOWN	UNKNOWN
DI-008	Sterilization	UNKNOWN — phages incompatible with gamma; aseptic processing or EtO unknown validation	UNKNOWN
DI-009	Phage K source	UNKNOWN — research-grade available; cGMP-grade requires manufacturing	UNKNOWN
DI-010	Electrospun Ti process	UNKNOWN achievable coating parameters on catheter geometry	UNKNOWN
DI-011	Shelf life (phage viability)	UNKNOWN — likely cold-chain dependent	UNKNOWN

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	Ti-coated catheter body with phage K layer	ABSENT	Ti coating process parameters; Phage immobilization chemistry; Phage K
DO-002	Electrospun Ti coating process specification	ABSENT	Electrospinning parameters; Adhesion to catheter substrate; Thickness
DO-003	Phage K immobilization process specification	ABSENT	Immobilization chemistry; Phage stock quality; QC release criteria
DO-004	Antimicrobial efficacy test protocol	CONCEPTUAL	

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
Phage K titer (initial)	UNKNOWN achievable surface density	PFU/cm ²	Coating process-dependent	Surface titer assay
Phage-bacteria adsorption rate k _{ads}	EXTERNAL_PRECEDENT (~10 ¹⁰ PFU/mL) (Published phage adsorption kinetics)	1/h	Published phage adsorption kinetics	Phage adsorption kinetics
Phage decay rate k _{decay}	UNKNOWN in CSF environment	1/h	UNKNOWN for chronic CSF exposure	Assay study in CSF mimic
Electrospun Ti coating thickness	UNKNOWN	μm	Electrospinning process parameters	SEM cross-section
Ti coating surface area multiplier	UNKNOWN — electrospinning increases surface area 10-1000x	g/g	Electrospinning process	BET surface area measurement
Bacterial challenge (worst case)	EXTERNAL_PRECEDENT (~10 ⁸ CFU/mL) (Published contamination literature)	CFU/mL	Published contamination literature	Repetitive challenge (in vitro)

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
Phage inactivation	Protein denaturation, desiccation, thermal degradation	Stabilization in storage buffer, periodic rehydration	Stability studies	UNKNOWN achievable half-life in CSF
Immobilization blocks receptor	Covalent coupling occludes phage receptor	Optimize coupling chemistry (e.g. phage receptor immobilization)	Phage adsorption assay	UNKNOWN achievable activity retention
Bacterial challenge exceeds capacity	High bacterial load overwhelms phage	Maximize phage density, combine with antibiotics	High challenge tests	UNKNOWN clinical contamination distribution
Host immune clearance of phage	Immune system inactivates phage	Phage surface modification to evade immune response	In vivo persistence studies	UNKNOWN immune response magnitude
Phage-resistant S. aureus	Bacteria evolve phage resistance	Phage cocktail (multiple phages)	Resistance evolution assay	UNKNOWN resistance rate in vivo
Ti coating delamination	Mechanical bond failure between phage and substrate	Surface activation, adhesion testing	ASTM G3359 + fatigue	UNKNOWN chronic adhesion

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
Phage inactivation over time	Protein denaturation, desiccation, chemical degradation	Stabilization matrix (trehalose) + spare DNA	UNKNOWN achievable half-life
Immobilization blocks phage adsorption	Covalent chemistry occludes receptor	Optimize coupling (orient receptor outward)	UNKNOWN achievable retention
Bacterial challenge exceeds capacity	High bacterial load overwhelms phage	Maximize Gamma_0 + antibiotics adjunct	UNKNOWN clinical contamination distribution
Host immune clearance of phage	Immune system inactivates phages	PEGylation or immunosuppressive coating	UNKNOWN immune response magnitude
Phage-resistant S. aureus	Bacteria evolve phage resistance	Phage cocktail (multiple phages)	UNKNOWN resistance rate in vivo
Ti coating delamination	Mechanical bond failure	Surface activation + adhesion testing	UNKNOWN chronic adhesion
Non-S. aureus pathogens	Phage K host range limited to S. aureus	Multiple phages (S. epidermidis coverage)	UNKNOWN coverage of full pathogen spectrum

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
V-001	Phage K surface density Gamma_0	Surface titer assay	Above model threshold	NOT_TESTED
V-002	Log reduction >= 3 vs S. aureus at 24h	ASTM E2180	Log reduction >= 3	NOT_TESTED
V-003	ISO 22196 antibacterial activity	ISO 22196	Pass	NOT_TESTED
V-004	Phage viability over lifetime	Aging in CSF mimic	Titer > 50% at target lifetime	NOT_TESTED
V-005	Biocompatibility ISO 10993	ISO 10993 series	Pass	NOT_TESTED

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VAL-001	Reduced infection rate in animal model	In vivo challenge study	Statistically significant reduction	NOT_PERFORMED
VAL-002	Reduced infection rate in patient cohort	Clinical trial (IDE)	Statistically significant reduction	NOT_PERFORMED

11. Materials

Component	Candidate Material	Source	Verification Required	Status
Catheter body	Silicone elastomer	Standard CSF shunt material	ISO 10993 + Ti coating adhesion	CANDIDATE
Antimicrobial coating	Electrospun titanium	Published Ti coating literature (POC 2009, 1900)	Coating process + adhesion + biocompatibility	CANDIDATE
Antimicrobial agent	S. aureus phage K	Published phage K literature	Host range confirmation + activity	CANDIDATE immunogenicity
Stabilization matrix	Trehalose	Standard phage lyoprotectant	Stability + biocompatibility	CANDIDATE

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
01	Catheter body (substrate) 1		CUSTOM_COMPONENT	UNKNOWN — silicone polyurethane (catheter)	Multiple	CRITICAL	Precedence bench + biocompatibility
02	Titanium target / precursor	Process-Dependent	SCHEMABLE	Titanium (Ti-6Al-4V or Ti-6Zr-4V)	Multiple (medical-grade)	CRITICAL	Coating QC + adhesion
03	S. aureus phage K stock (P40)	100 µg	COM_BIOL	Phage particles in buffer	UNKNOWN — cGMP manufacturer	CRITICAL	Precedence + sterility + endotoxin
04	Stabilization matrix (trehalose)	Process-Dependent	SCHEMABLE	Trehalose or sucrose	Multiple (Sigma, etc.)	HIGH	Stability study

13. Manufacturing

Status: ENGINEERING_CANDIDATE — no process validated for phage-active catheter

Candidate Processes:

- Electrospinning of Ti coating on catheter: Published electrospinning literature - Custom process development for cylindrical catheter geometry
- Phage K propagation and purification: Standard phage manufacturing - cGMP manufacturer required for clinical-grade
- Phage immobilization on Ti surface: Standard phage immobilization chemistry - Custom process development required
- Aseptic assembly (terminal sterilization likely incompatible with phage): Standard aseptic processing - Phage viability under EtO UNKNOWN

14. External Evidence

Source 1: Preventing *S. aureus* biofilm formation on titanium surfaces by

URL: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6693497>

Excerpt: by ALR López · 2019 · Cited by 72 — β -peptide-containing films fabricated on rough titanium substrate surfaces can release sufficient quantities of β -peptide to prevent *S. aureus* biofilm ...

Source 2: *Staphylococcus aureus* resistance on titanium coated with

URL: <https://www.sciencedirect.com/science/article/abs/pii/S0142961210010537>

Excerpt: by X Khoo · 2010 · Cited by 79 — TBP4-PEG forms a serum-resistant surface coating capable of preventing *S. aureus* colonization and subsequent biofilm formation.

Source 3: Recent developments in preventing catheter-related infections

URL: <https://pubs.aip.org/aip/bmf/article/18/5/051506/3316469/Recent-developments-in-preventing-catheter-related>

Excerpt: Oct 11, 2024 — Under static conditions, PU catheters coated with oil-infused PDMS surface layers inhibited *E. coli* and *S. aureus* biofilm formation.

Source 4: Antibiotic-Free Solutions for the Development of Biofilm

URL: https://link.springer.com/chapter/10.1007/978-3-031-04484-7_21

Excerpt: by B Costa · 2022 · Cited by 1 — All coatings showed antibacterial and antifungal properties, *Staphylococcus aureus* (MRSA)

Source 5: Bacteriophage-Mediated Control of Biofilm: A Promising

URL: <https://www.frontiersin.org/journals/microbiology/articles/10.3389/fmicb.2022.825828/full>

Excerpt: by C Chang · 2022 · Cited by 177 — catheter-associated biofilm clearance may involve phage is used as a gel-like coating on the catheter to reduce the bacterial adhesion to the ...

15. Transfer Boundary

YOU RECEIVE:

- + Phage K + Ti coating concept + mechanism description
- + Phage adsorption + Monod biofilm governing equations
- + Bench test protocol (ASTM E2180, ISO 22196, titer assay)
- + External precedent catalog (phage K, Ti coatings)
- + Critical UNKNOWN disclosure (phage stability, immune response, resistance)
- + This engineering dossier with failure analysis + build plan

YOU MUST DEVELOP:

- > cGMP-grade phage K manufacturing process
- > Electrospun Ti coating process for catheter geometry
- > Phage immobilization chemistry + QC specs
- > Aseptic assembly process (terminal sterilization likely incompatible)
- > Regulatory submission (likely combination product — biologics + device; complex pathway)
- > Clinical validation (infection rate endpoint)
- > Long-term stability + resistance evolution monitoring

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- Achievable phage surface density Gamma_0 on electrospun Ti coating — UNKNOWN
- Phage K half-life in CSF environment at 37°C — UNKNOWN
- Infectivity retention after immobilization chemistry — UNKNOWN
- In vivo phage persistence and immune response — UNKNOWN
- Phage resistance evolution rate in clinical use — UNKNOWN
- Regulatory pathway for phage-containing device — UNKNOWN (FDA phage guidance evolving; likely complex combination product)
- Manufacturing scale-up for phage-coated catheter — UNKNOWN
- Clinical endpoint (infection rate reduction magnitude) — UNKNOWN until trial

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	Ti-coated catheter coupon (spin-coated)	Electrospinning setup	SEM, thickness, coating area	Coating uniformity, adherence	Coated catheter coupon	2 weeks
WP-02	Phage K stock (research grade)	Phage propagation, purification	PFU/ml assay	Therapeutic target; host range	Phage K stock	2 weeks
WP-03	Phage-coated Ti coupon	Surface titer assay, infectivity assay	Infectivity assay, surface area	Calculated infectivity reduction	Infectivity reduction	2 weeks

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-04	Phage-coated catheter	ASTM F2180 test setup	Seal integrity at 24h	Log reduction >= 3	Antimicrobial efficacy report	2 weeks
WP-05	Phage-coated specimen	Aging chamber	Titer 3556 vs time	Titer > 50% at target	Stability report	12-24 weeks (chronic)
WP-06	Biocompatibility specimen	ISO 10993 catheter	ISO 10993 seripage	ISO 10993 pass	ISO 10993 report	12 weeks (external lab)
WP-07	Animal model contamination	Animal challenge	Microbiological rate vs control	Statistically significant	Reduction efficacy report	6 weeks

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 4 governing equations and a 7-step build plan. The next experiment is clearly specified. Target buyer: Anti-infection medical device company; neuroshunt manufacturer.

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: Phage K cannot retain infectivity when immobilized on Ti surface for >24h

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: Phage K cannot retain infectivity when immobilized on Ti surface for >24h

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.